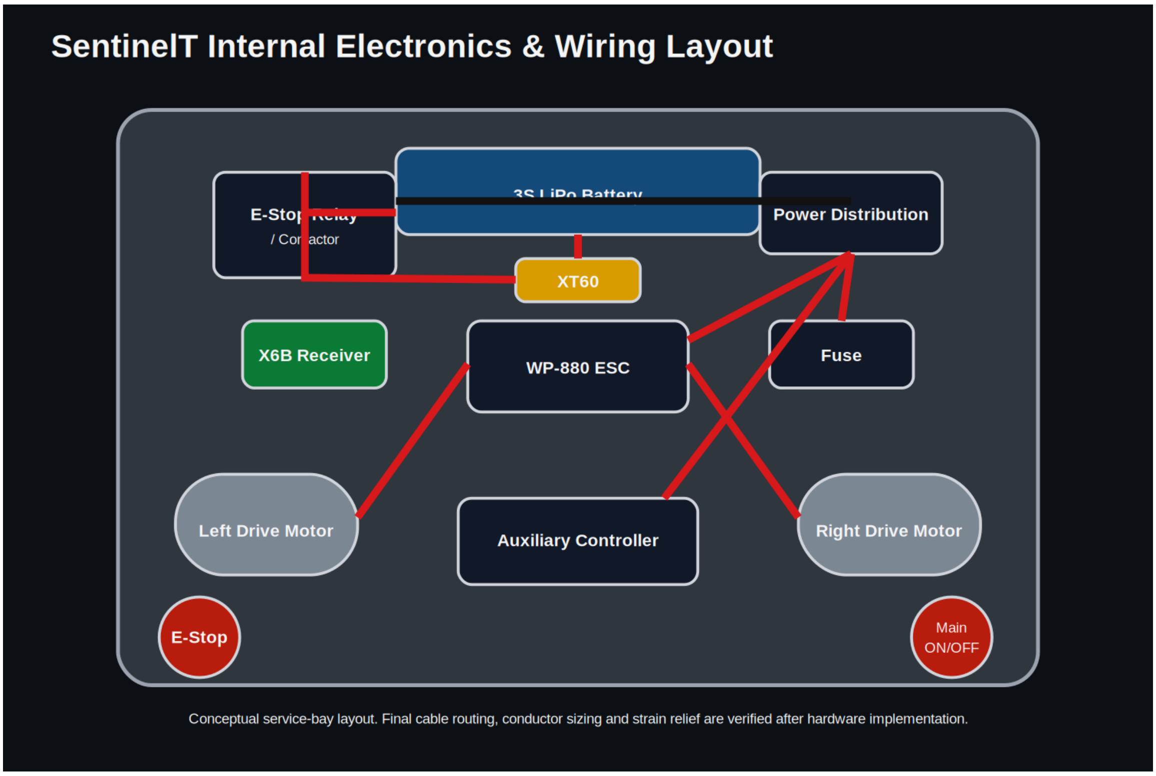
SENTINELT

Electronic Design

Project Developer: TG Assegaai



Electronic architecture, wiring, circuit safety and E-Stop integration

1. Electronic Architecture

SentinelT uses a 3S DC power system with protected high-current distribution, a 2.4 GHz radio-control link, brushed drive control and a separate auxiliary branch. The accessible main ON/OFF isolator and the E-Stop-controlled high-current isolation stage are upstream of the drive and auxiliary actuator buses.

Function	Hardware / method
RC link	FlySky FS-i6X + X6B receiver, 2.4 GHz
Battery	3S 11.1 V LiPo with XT60
Drive ESC	HobbyWing QuicRun WP-880 dual brushed ESC
Drive motors	Two 12 V brushed 37D gearmotors
Protection	Main fuse / fused distribution
Main power control	Accessible battery isolator
Emergency stop	Mushroom E-Stop controlling a high-current relay/contactor
Auxiliary branch	Guarded active-mechanism controller/actuator; finalized after hardware matching

2. Circuit Schematic and Power Flow

SentinelT Power & E-Stop Architecture

2.4 GHz RC | Main ON/OFF | Independent emergency power-disable path

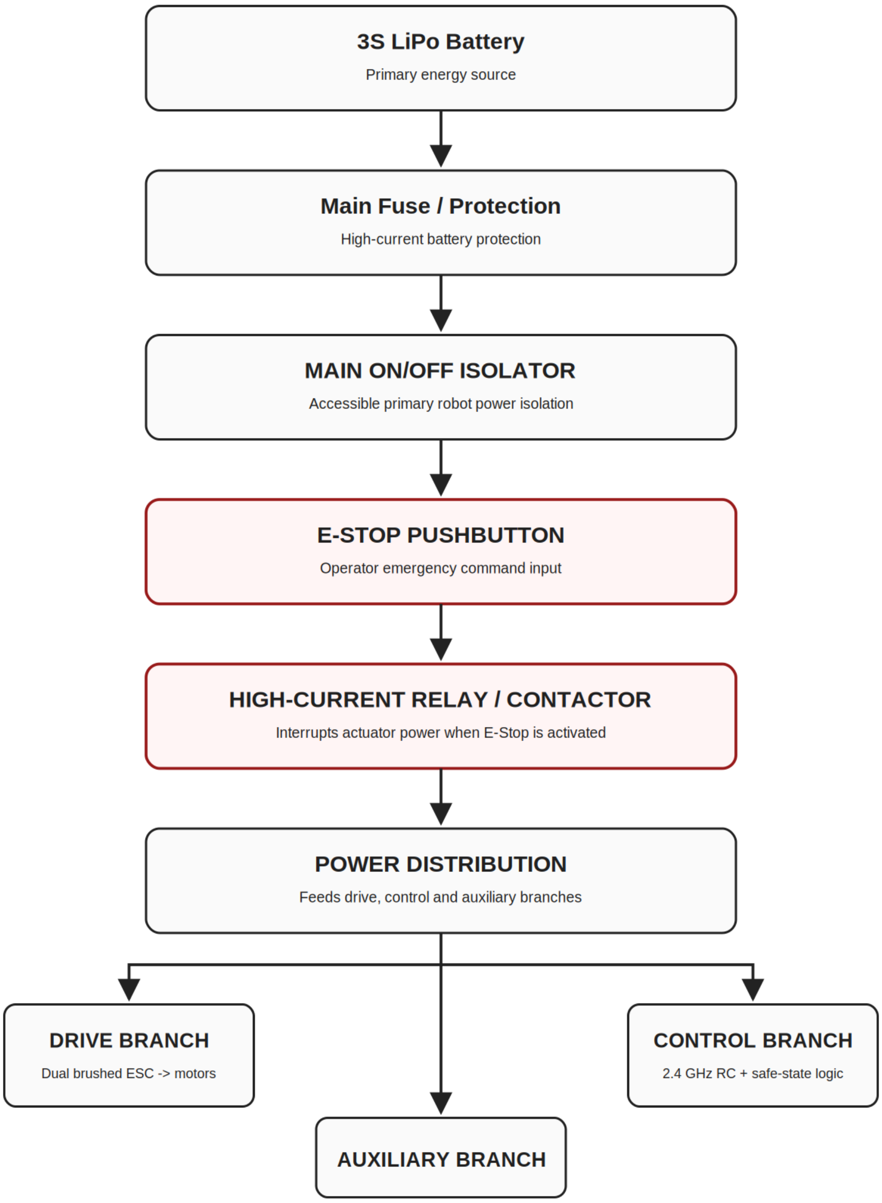


Figure 1. SentinelT power and control architecture.

3. E-Stop Integration

The E-Stop is integrated as a hardware power-disable function. The emergency pushbutton opens the relay/contactor control loop. The contactor main contacts then interrupt the actuator supply. This design does not depend on the microcontroller or RC software to remove motor power.

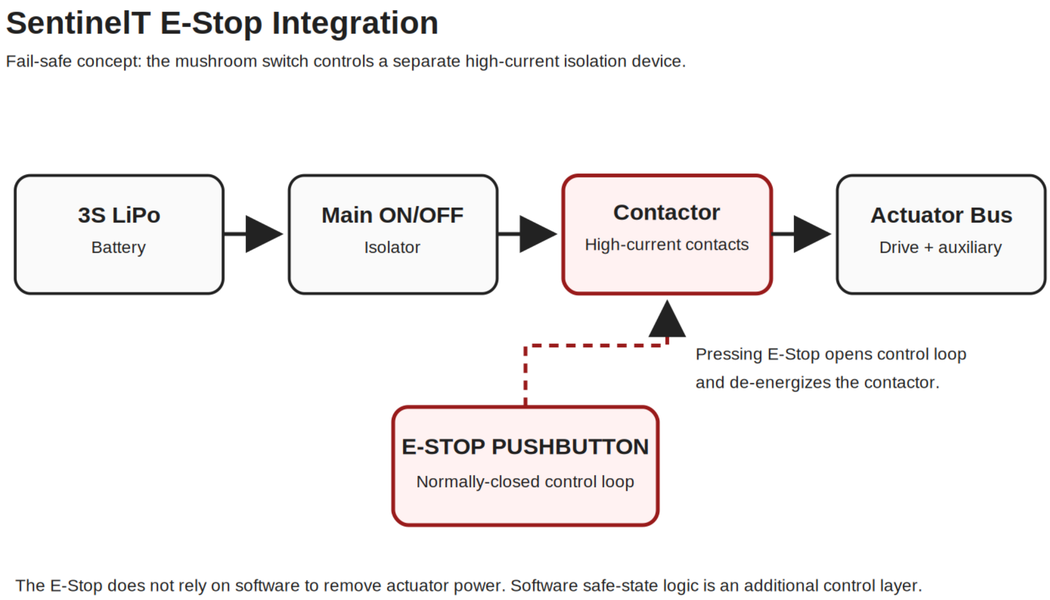


Figure 2. Mandatory E-Stop integration.

The main ON/OFF switch remains a separate accessible isolation device. When hardware is implemented, the selected relay/contactor, fuse, conductor size and connector ratings will be verified against measured current.

4. Wiring and Internal Placement

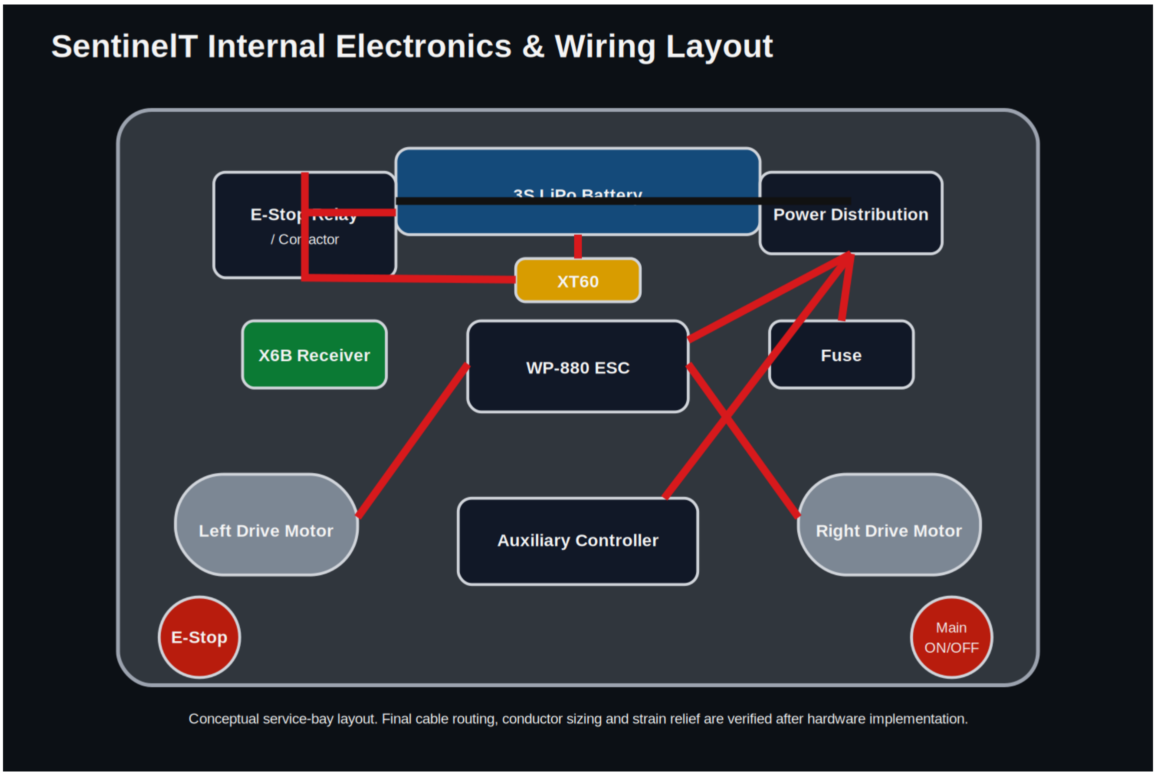


Figure 3. Internal wiring/component layout.

High-current battery/actuator wiring is separated conceptually from low-current RC/control wiring. Final routing, strain relief and wire sizing will be verified after hardware implementation.

5. Component Justification

Component	Reason for selection
3S LiPo	Matches the 12 V-class drivetrain and current power architecture
WP-880 dual brushed ESC	Provides brushed drivetrain control in one protected controller package
FS-i6X + X6B	2.4 GHz human-operated RC link with multiple channels
37D gearmotors	Compact geared drive units with 6 mm shafts
Main isolator	Provides accessible primary ON/OFF control
E-Stop + relay/contactor	Provides independent actuator power cut
Fuse / distribution	Protects and organizes the high-current battery path

Project Developer: TG Assegaai

Signature: Thato Glen Assegaai